

PHYSICAL AI ENGINEERING

Make the digital twin accountable to the real operation.

July 15, 2026

EY Consulting Hiring Team

ROLE FIT AT A GLANCE

Digital twin + automation

Warehouse robotics, pick/pack/stow, conveyance and material-flow decision support.

Embodied systems

Machine vision, regulated autonomous operations and computer-vision concepts.

High-reliability delivery

Advanced electronics, quality systems, root cause and high-consequence programs.

Scaled operations

Amazon launch discipline, operating cadence, throughput visibility and customer outcomes.

FIRST CONTRIBUTION

Help the team carry one coherent evidence case from simulation through physical validation, client authority, deployment and sustainment.

90-DAY ORIENTATION

- Read EY methods, alliances, authorities and active workstreams.
- Lead one bounded technical seam with explicit acceptance evidence.
- Transfer ownership and harvest reusable learning without abstracting away client reality.

EY's Physical AI Engineering Manager role is compelling because it is not merely a robotics job. It asks someone to convert simulation, sensing, machine learning and software delivery into client-ready capability - while leading a technical workstream, developing people, protecting engagement economics and building trust with the operators who will inherit the system.

That is the operating seam I have worked across throughout my career. I bring digital-twin experience related to warehouse automation, robotic pick/pack/stow, conveyance and material-flow decision support, together with current leadership in a software-enabled robotics and machine-vision environment. At Vape-Jet, I connect production, engineering, quality, support, telemetry and enterprise workflows around a fielded fleet of approximately 1,000 robots. The work is making machine and workflow evidence visible enough to prioritize, escalate, correct and sustain.

My earlier work adds the consequence discipline this role requires. At ZeusVu, I led regulated autonomous-drone operations with a 100% safety record and no FAA incidents, and developed TensorFlow-based concepts for aerial inventory and field intelligence. At Compunetics, I led engineering, manufacturing, quality and complex technical programs in advanced electronics and high-reliability environments, collaborating with organizations including DoD, CERN, Intel and AMD. At Amazon, I built operating mechanisms for a 24/7 launch environment supporting approximately five million second-day orders annually.

I would enter EY with a strong physical-systems and digital-twin foundation, a manager's accountability for workstream quality and economics, and the humility to learn the specific platforms and methods used by EY teams. ROS, Omniverse, MLflow and related tooling are important implementation environments; they are not substitutes for defining evidence, authority, risk, run-state ownership and client adoption. My contribution would be to help specialists and clients keep those elements connected.

The accompanying candidate vision presents a Reality Transfer Case for carrying evidence from the virtual model into physical validation and client operation. It also includes a 90-day entry plan and executive questions designed to test where EY most needs leverage. The work is available at <https://russelldudek.github.io/EY/>.

I would welcome the opportunity to discuss how my digital-twin, robotics, high-reliability and operating experience could help EY turn physical-AI engagements into capabilities clients can trust, operate and extend.

My value is the seam: digital-twin logic, physical evidence, client decisions and sustained operating ownership.

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DIGITAL TWINS | ROBOTICS | MACHINE VISION | HIGH-RELIABILITY DELIVERY

Candidate vision and application artifacts

<https://russelldudek.github.io/EY/>